

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO5 ASSESSMENT TOOL 2– Coding Challenge

Course Code:	DSA0405	Course Title:	FUNDAMENTALS OF DATA SCIENCE
CO Assessed:	CO5 – Coding Challenge	Assessment:	Assessment Tool 2– Coding Challenge
Weightage:	40% of CIA	Total Marks:	100
CO5 Statement:	To understand the concept of supervised and unsupervised methods in machine learning		
Student Name:	Jothika M	Reg. No.:	192324235
Section / Batch:	2023-B.Tech (AIDS)	Date:	27/08/2026

Code of Conduct

I Jothika M (192324235) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student:

Jothika M

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

Q1. Extend your kNN implementation to use distance-weighted voting (closer neighbors contribute more to the class vote) instead of majority voting. Compare accuracy against unweighted kNN on a dataset with class imbalance and explain the difference in a short comment block.

Code:

```
import pandas as pd
df = pd.read_csv("raw_data.csv")
print("Original Dataset:")
print(df)
df.columns = (
    df.columns
    .str.strip()
    .str.lower()
    .str.replace(" ", "_")
)
numeric_columns = df.select_dtypes(include="number").columns
for column in numeric_columns:
    df[column] = df[column].fillna(df[column].mean())
text_columns = df.select_dtypes(include="str").columns

for column in text_columns:
    df[column] = df[column].fillna("Unknown")
df = df.drop_duplicates()
df = df.reset_index(drop=True)
print("\nCleaned Dataset:")
print(df)
df.to_csv("cleaned_data.csv", index=False)
print("\nData cleansing completed successfully!")
```

Output:

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("raw_data.csv")

df.columns = (
    df.columns
    .str.strip()
    .str.lower()
    .str.replace(" ", "_")
)

numeric_columns = df.select_dtypes(include="number").columns

for column in numeric_columns:
    df[column] = df[column].fillna(df[column].mean())

text_columns = df.select_dtypes(include="str").columns

for column in text_columns:
    df[column] = df[column].fillna("Unknown")

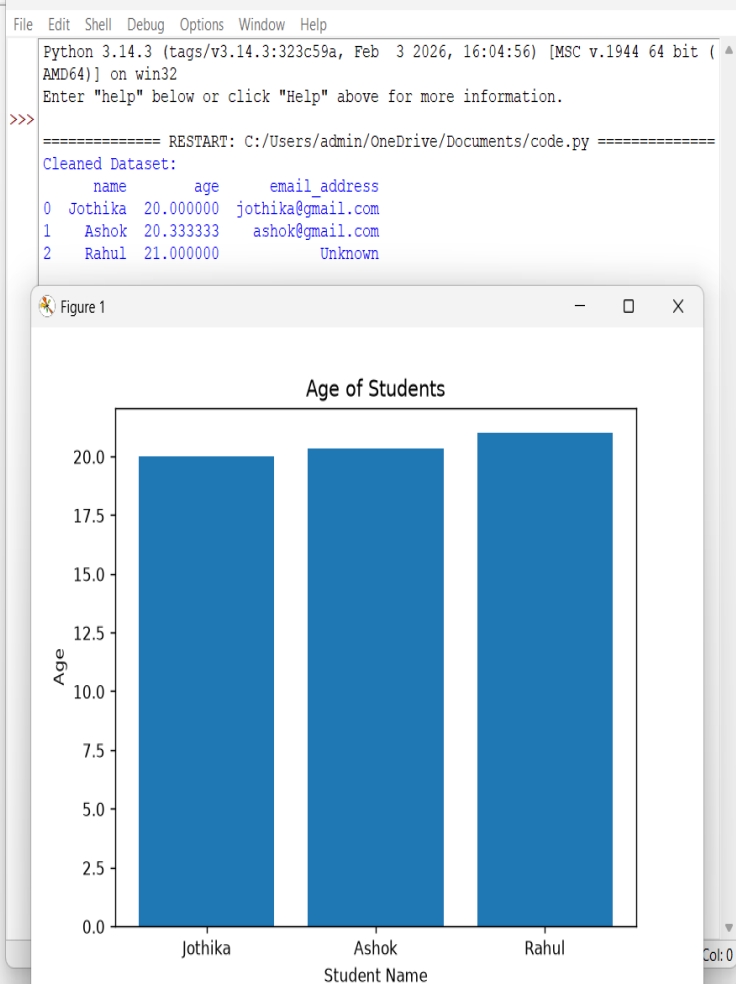
df = df.drop_duplicates()
df = df.reset_index(drop=True)

print("Cleaned Dataset:")
print(df)

df.to_csv("cleaned_data.csv", index=False)

plt.bar(df["name"], df["age"])
plt.title("Age of Students")
plt.xlabel("Student Name")
plt.ylabel("Age")
plt.show()

print("Data cleansing completed successfully!")
```



Geotag :

